Daily Habit Tracker with Streaks

Semester Project Report by

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Abstract

This project implements a simple client-side login interface using React and React Router DOM. The application consists of a Login component that provides a user-friendly interface built with Bootstrap for styling. It includes form handling, state management using React Hooks (useState), and navigation using useNavigate from react-router-dom. The login logic validates user input against predefined credentials (user@example.com and 123456). Upon successful authentication, the user is navigated to the dashboard route. If authentication fails, an appropriate error message is displayed. The project entry point (index.js) initializes the React application using ReactDOM.createRoot and wraps the app with React.StrictMode. To support navigation, it is recommended to wrap the App component with BrowserRouter. While this application does not connect to a backend or use persistent storage, it effectively demonstrates the principles of a front-end login workflow While this application does not connect to a backend or use persistent storage, it effectively demonstrates the principles of a front-end login workflow. This project demonstrates foundational React concepts, including controlled components, routing, conditional rendering for error handling, and form validation, making it a suitable example for learning basic authentication flow in React.

Introduction

In recent years, significant efforts have been made to develop intelligent and interactive systems that help users monitor and improve their personal productivity and lifestyle. Among these, habit tracking applications have emerged as a powerful tool to support behavior change by allowing users to record and maintain their daily routines digitally. Traditionally, users have relied on physical journals or reminders to track habits; however, with the advent of web-based and mobile technologies, digital habit trackers offer greater flexibility, personalization, and real-time analytics. Modern habit trackers often incorporate motivational elements such as streaks, where users are rewarded for maintaining consistency over consecutive days.

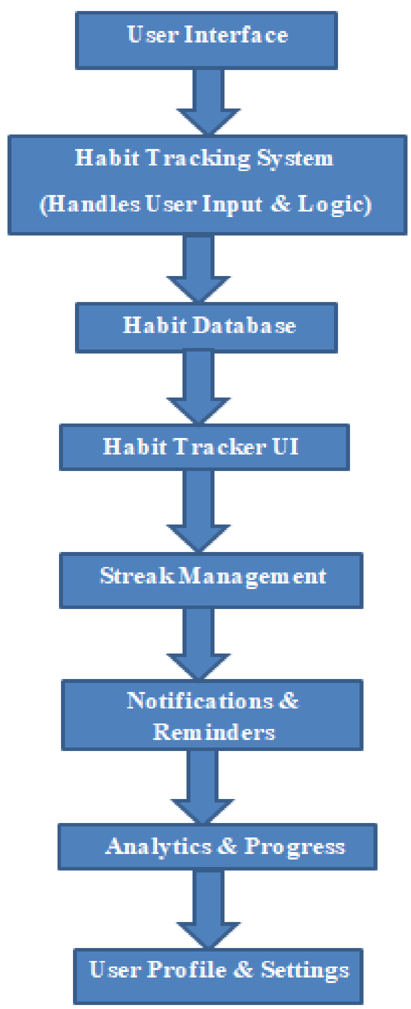

Figure 1: Block Diagram of Daily Habit Tracker with Streaks

This method reinforces positive behavior through gamification, making users more engaged and accountable. A daily habit tracker with a streak system removes various psychological barriers to habit formation by providing visual feedback, reminders, and encouragement. Moreover, by utilizing user-friendly technologies like React.js and Bootstrap, such systems ensure that habit tracking is not only efficient but also intuitive and visually appealing. The focus of this project is to create a responsive and dynamic habit tracking system that enables users to add, monitor, and manage their daily habits while encouraging consistency through streak tracking. The application represents a modern approach to self-improvement, blending productivity with user experience design. Through gamification elements such as streak rewards, the system fosters a sense of accomplishment, helping individuals stay consistent in their self-improvement journey. Psychologists suggest that habit formation is driven by both external motivation and internal rewards, and streaks provide just that: an external motivation to keep going, alongside an intrinsic sense of accomplishment as users see their progress reflected in their streaks. A user may initially be motivated to create a new habit for personal growth or productivity, but as their streaks grow longer, their desire to maintain their streak becomes an even greater motivator. The application will allow users to create a personalized habit list, set daily goals, and track their progress. The streaks will be visually represented to encourage users to keep up with their habits. By logging their progress daily, users can receive motivating reminders and rewards for their consistent effort, thereby enhancing their productivity, improving their well-being, and fostering positive long-term change.

Problem Statement

The problem is to develop a **Daily Habit Tracker with Streaks** that helps individuals stay consistent with their daily habits by tracking their progress, providing visual streaks to motivate continued success, and sending timely reminders. The system should allow users to create and manage habits, visualize their progress through streaks (consecutive days of completion), and receive notifications to encourage habit consistency. Additionally, the application should offer personalized user profiles, track completion rates, and provide insights to improve habit formation, all while being available across multiple platforms for seamless access.

Project aims and goals

The Daily Habit Tracker with Streaks project aims to create an intuitive application that helps users build and maintain positive habits by allowing them to track their daily activities and monitor their progress. The core goal is to provide a user-friendly platform where individuals can easily create, manage, and track their habits, while the streak feature motivates them to maintain consistency by tracking consecutive days of completion. With personalized reminders, progress visualization, and cross-platform accessibility, the application is designed to encourage users to stay committed to their goals and achieve personal growth through continuous habit development. The project seeks to foster long-term habit formation by offering a motivating and engaging user experience.

Project Methodology

The development process follows the standard software development life cycle (SDLC), with an emphasis on iterative development, testing, and user feedback integration.

Requirement Analysis

The first step in the methodology involves gathering and analyzing the requirements for the application. This phase includes understanding the needs of the target users, which are individuals seeking to track and maintain their daily habits. Through user interviews and surveys, the core functionalities of the application, such as habit creation, tracking, progress visualization, and streak management, were identified. Non-functional requirements, including performance, usability, and security, were also outlined to ensure the application is both efficient and secure for users.

Design

Based on the requirements, the system architecture and user interface (UI) are designed. The design phase focuses on creating wireframes and flowcharts to outline how users will interact with the app. User interface design principles are followed to ensure the app is intuitive and easy to navigate.

Development

React.js was used to create the interactive components of the application. Users can add, track, and visualize their habits through a simple interface. Features like habit lists, streak counters, and progress graphs were implemented to help users easily track their daily habits and maintain their streaks.

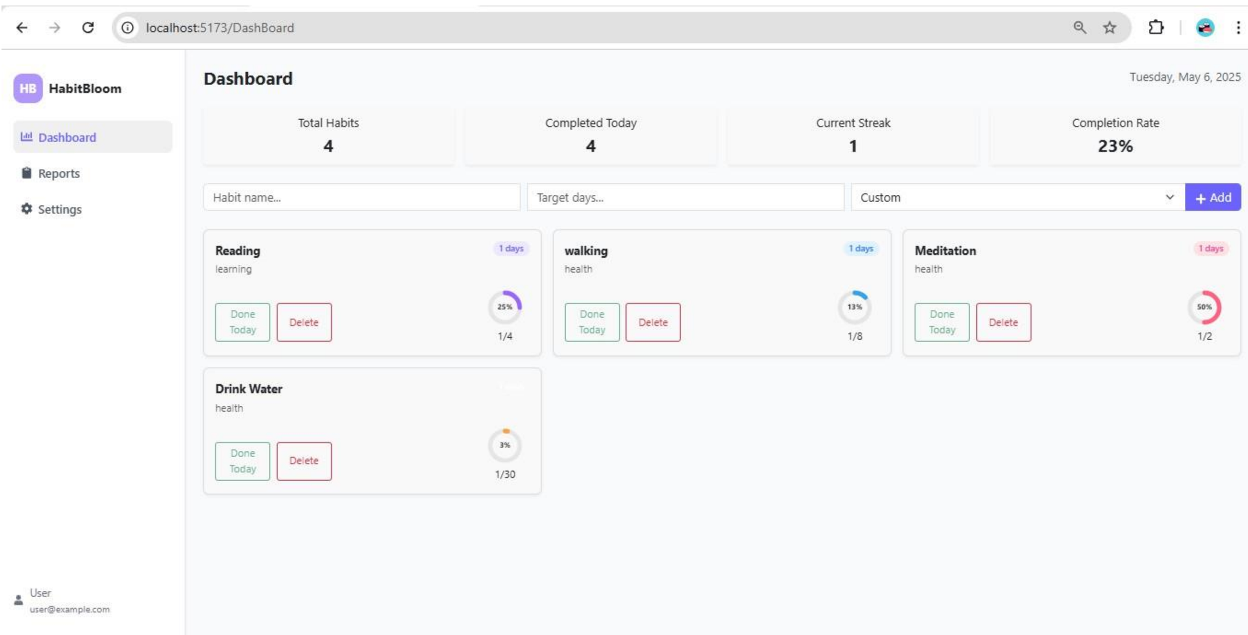

Figure 2: Habit tracking dashboard and overview.

Testing

Each function and component was tested individually to ensure it operates correctly. This included testing streak calculation logic, habit tracking functionalities, and user registration/authentication.

Maintenance and Updates

The application was continuously monitored for performance issues and bugs. Users provided feedback, which was used to improve features and fix any issues. Regular updates were planned to add new features such as additional habit types, social sharing options, and more detailed progress reports.

- **Bug Fixes**: Any bugs reported by users or discovered through monitoring were fixed in a timely manner.
- **Feature Enhancements**: Future updates will include additional features such as reminder notifications, multi-user support, and habit analysis with AI-based recommendations.

Conclusion

The Daily Habit Tracker with Streaks application successfully addresses the need for a simple yet effective tool that helps individuals build and maintain positive habits over time. By leveraging modern web technologies such as React.js, the application offers a responsive, interactive, and user-friendly experience. The inclusion of streak tracking serves as a motivational feature, encouraging users to stay consistent with their daily routines. Through thoughtful design, robust functionality, and continuous feedback-driven improvements, the project demonstrates how digital tools can positively influence personal development and productivity. Ultimately, this project not only fulfills its initial objectives but also lays the foundation for future enhancements such as reminder notifications, habit analytics, and social sharing, making it a scalable and impactful solution for habit formation and self-discipline.